

HW # 36

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CS-F

Prob 1

Self-adjoint: $A^* = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ Yes

Positive Semi-definite: $x^* A x$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 - x_2 \\ -x_1 + x_2 \end{bmatrix}$$

$$= x_1^2 - x_1 x_2 - x_1 x_2 + x_2^2$$

$$= (x_1 - x_2)^2 \geq 0 \quad \text{Yes}$$

Positive Eigenvalues: $\begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} = 0$

$$(1-\lambda)^2 - 1 = 0$$

$$1 + \lambda^2 - 2\lambda - 1 = 0$$

$$\lambda = 0 \quad \text{or} \quad \lambda = 2 \quad \text{Yes}$$

$$A = B^* B$$

let,

$B \Rightarrow$ obtained by diagonalization of symmetric matrices Eigenvalue Decomposition

$$B = P \sqrt{D}$$

Prob 2:-

$$A = \begin{bmatrix} x & 4 \\ 4 & y \end{bmatrix}$$

for A to be positive definite

$$\cancel{xy \neq 16} \quad xy > 16$$

let $x = 2$, $y = 9$

$$A = \begin{bmatrix} 2 & 4 \\ 4 & 9 \end{bmatrix}$$

$$|2| = 2$$

$$\begin{vmatrix} 2 & 4 \\ 4 & 9 \end{vmatrix} = 18 - 16 = 2 > 0$$

Therefore, it is +ive definit
by Sylvester's Criterion.

Prob 3:-

$$A = \begin{bmatrix} 2 & -1 & i \\ -1 & 2 & 1 \\ -i & 1 & 2 \end{bmatrix}$$

$$|2| = 2 > 0$$

$$\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3 > 0$$

$$\begin{aligned} \begin{vmatrix} 2 & -1 & i \\ -1 & 2 & 1 \\ -i & 1 & 2 \end{vmatrix} &= 2(3) + 1(-2+i) + i(-1+2i) \\ &= 6 - 2 + i - i - 2i^2 \\ &= 6 - 2 - 2 \\ &= 2 > 0 \end{aligned}$$

Positive Definite.

Problem 4:

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad \begin{array}{l} a_{11}: 1 \geq 0 \\ a_{22}: 1 \geq 0 \\ \det(A): 0 \geq 0 \end{array} \quad \text{Semi definite}$$

$$B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \quad \begin{array}{l} a_{11}: 1 \geq 0 \\ a_{22}: 1 \geq 0 \\ \det(A): -3 < 0 \end{array} \quad \text{Not semi-Positive}$$

Problem 5:

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \quad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$$

→ A is Symmetric → A self-adjoint exists

$$\rightarrow \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(x_1 - x_2)^2 \geq 0 \quad \text{semi-definite.}$$

$$\rightarrow \begin{vmatrix} 1-\lambda & 2 \\ 2 & 5-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(5-\lambda) - 4 = 0$$

$$5 - 6\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 6\lambda + 1 = 0$$

$$\lambda = 3 + 2\sqrt{2} \quad \lambda = 3 - 2\sqrt{2} \geq 0 \quad \text{Positive.}$$

Since A is a Positive definite matrix
by Theorem 6

$$\langle v, w \rangle = v^* A w$$

represents an inner product.